

Question Number	Answer	Additional guidance	Mark
6(a)(i)	An answer that includes the following point: • amyloplast (1)		(1)

Question Number	Answer	Additional guidance	Mark
6(a)(ii)	An explanation that includes three of the following points: • {polymer of / contains} glucose because (glucose) is an energy source (1) • {branched / 1,6 bonds} therefore can be rapidly hydrolysed (to release glucose/energy) (1) • compact so more {glucose/energy} can be stored / compact so high energy density} (1) • insoluble therefore does not affect osmosis (1)	ACCEPT branched so quickly broken down ignore easy to break down ACCEPT compact so does not take up much space ACCEPT too big to diffuse across cell surface membrane / large molecules therefore remain in cells	(3)

Question Number	Answer	Mark
6(b)(i)	The only correct answer is D 67 % <i>A is not correct because that is 100-67</i> <i>B is not correct because the correct answer is 67 %</i> <i>C is not correct because that is incorrectly rounded from 66.6667</i>	(1)

Question Number	Answer	Additional guidance	Mark
6(b)(ii)	1074 (million tonnes)		(1)

Question Number	Answer	
6 (c)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. • both types of cassava require less water (than maize) / maize needs {more / lots of} water • both types of cassava require fewer nutrients (than maize) / maize needs {more / lots of} nutrients • both types of cassava contain more carbohydrates (than maize) / maize contains less carbohydrates • maize contains more protein / both types of cassava contain less protein (than maize) • higher mean dry mass for bitter cassava / lower mean dry mass for sweet cassava • higher concentrations of {(bitter) chemicals / hydrogen cyanide} in bitter cassava / sweet cassava contains more (bitter) chemicals than maize • if not processed properly then {(bitter) chemicals / hydrogen cyanide} will remain in the crop and cause death • higher mean dry mass means more food is grown / growing bitter cassava feeds more people • carbohydrates are needed as a source of energy / broken down into glucose / needed for respiration • more maize would need to be eaten (to meet body need) • protein is needed for {growth and repair / mitosis / making new proteins, hormones} etc • reduced growth with low protein diet / more cassava would need to be eaten / converse • reduced cost to grow cassava than maize / less irrigation needed / fewer fertilisers needed / converse • ability to grow cassava in more areas (than maize) • reduced {cost/labour} to grow maize and sweet cassava (than bitter cassava) as no need to remove (bitter) chemicals / converse • higher concentrations of {(bitter) chemicals / hydrogen cyanide} will result in {fewer pesticides being needed by farmers / increased yields for farmers} / converse for {maize/sweet cassava} • as {(bitter) chemicals / hydrogen cyanide} deter {herbivores / insects / pests} that feed on the plant	(6)

			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	An explanation may be attempted but with limited interpretation or analysis of the scientific information and with a focus on mainly just one piece of scientific information. The explanation will contain basic information, with some attempt made to link knowledge and understanding to the given context.	consideration of advantage(s) and disadvantage(s) from given information only
Level 2	3-4	An explanation will be given, with occasional evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows some linkages and lines of scientific reasoning, with some structure.	all level 1 including two of the following aspects -explanation of the {advantage of higher /disadvantage of lower} carbohydrate content - explanation of the {advantage of higher /disadvantage of lower} protein content -consideration of disadvantage of cost of providing resources / removing bitter chemicals -consideration of advantage of {not having to buy pesticides /reduced loss of crops to pests} / converse for maize and sweet cassava
Level 3	5-6	An explanation is made that is supported throughout by sustained application of relevant evidence of analysis, interpretation and/or evaluation of both pieces of scientific information. The explanation shows a well-developed and sustained line of scientific reasoning, which is clear and logically structured.	all level 2 including all the following aspects -explanation of the {advantage of higher /disadvantage of lower} carbohydrate content - explanation of the {advantage of higher /disadvantage of lower} protein content -consideration of disadvantage of cost of providing resources / removing bitter chemicals -consideration of advantage of {not having to buy pesticides /reduced loss of crops to pests} / converse for maize and sweet cassava